# Maxima and Minima of functions of one variable

Let c be a point in a domain D of a function f. Then f(c) is the

⇒ absolute maximum value of f on D if $f(c) \geq f(x)$ for all x in D.

⇒ absolute minimum value of f on D if $f(c) \leq f(x)$ for all x in D.

## Definition:

Let c be a point in a domain D of a function f. Then f(c) is the

⇒ local maximum value of f if $f(c) \geq f(x)$ when x is near c.

⇒ local minimum value of f if $f(c) \leq f(x)$ when x is near c.

## Critical Point:

A critical point of a function f is a point c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exists.

If f has local maximum value or minimum value at c, then c is a critical point of f.

**Example:**

**Find the critical points of the following functions**

**(i)** $\boldsymbol{f(x) = x^3 + x^2 - x}$

**(ii)** $\boldsymbol{f(x) = x^{\frac{5}{4}} - 2x^{\frac{1}{4}}}$

**Solutions:**

**(i)** $\boldsymbol{f(x) = x^3 + x^2 - x}$

$$f'(x) = 3x^2 + 2x - 1$$

$$f'(x) = 0 \Rightarrow 3x^2 + 2x - 1 = 0$$

$$\Rightarrow (3x - 1)(x + 1) = 0$$

$$\Rightarrow x = \frac{1}{3}\ , -1$$

Critical points are $x = \frac{1}{3}\ , -1$.

**(ii)** $\boldsymbol{f(x) = x^{\frac{5}{4}} - 2x^{\frac{1}{4}}}$

$$f'(x) = \frac{5}{4}x^{\frac{1}{4}} - \frac{1}{4}2x^{-\frac{3}{4}}$$

$$f'(x) = 0 \Rightarrow \frac{1}{4}x^{\frac{1}{4}}(5 - 2x^{-1}) = 0$$

$$\Rightarrow \frac{1}{4}x^{\frac{1}{4}} = 0\ ,\ (5 - 2x^{-1}) = 0$$

$$\Rightarrow x = 0\ ,\ \frac{5x-2}{x} = 0$$

$\Rightarrow x = \frac{2}{5}$

Critical points are $x = 0 , \frac{2}{5}$.

**Example:**

**Find the absolute maximum and absolute minimum of**

(i) $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$on [-2 , 3]

(ii) $f(x) = x - 2sinx$ on [0 , $2\pi$]

(iii) $(x) = x - logx$ on [ $\frac{1}{2}$ , 2]

**Solutions:**

**(i) $f(x) = 3x^4 - 4x^3 - 12x^2 + 1$**

$f(x) = 3x^4 - 4x^3 - 12x^2 + 1$ is continuous on [-2 , 3]

$f'(x) = 12x^3 - 12x^2 - 24x$

$f'(x) = 0 \Rightarrow 12x^3 - 12x^2 - 24x = 0$

$\Rightarrow x(x+1)(x-2) = 0$

$\Rightarrow x = 0 , -1 , 2$ are the critical points.

The values of $f(x)$ at critical points are

$f(0) = 3(0^4) - 4(0^3) - 12(0^2) + 1 = 1$

$f(-1) = 3(-1)^4 - 4(-1)^3 - 12(-1)^2 + 1$

$= 3 + 4 - 12 + 1 = -4$

$f(2) = 3(2)^4 - 4(2)^3 - 12(2)^2 + 1$

$= 48 - 32 - 48 + 1 = -31$

The value of $f(x)$ at the end points of the interval are

$f(-2) = 3(-2)^4 - 4(-2)^3 - 12(-2)^2 + 1$

$= 48 + 32 - 48 + 1 = 33$

$f(3) = 3(3)^4 - 4(3)^3 - 12(3)^2 + 1$

$= 243 - 112 - 108 + 1 = 28$

Absolute minimum value is $f(2) = -31$

Absolute maximum value is $f(-2) = 33$

**(ii) $f$(iv) $f(x) = x - 2sinx$ on [0 , $2\pi$]**

**Solution:**

$f(x) = x - 2sinx$ is continuous on [0 , $2\pi$]

$$f'(x) = 1 - 2cosx$$

$$f'(x) = 0 \Rightarrow 1 - 2cosx = 0$$

$$\Rightarrow cosx = \frac{1}{2}$$

$$\Rightarrow x = cos^{-1}\left(\frac{1}{2}\right)$$

$\Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$ are the critical points.

The values of f(x) at critical points are

$$f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - 2sin\frac{\pi}{3}$$

$$= \frac{\pi}{3} - 2\frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{3} - \sqrt{3} \approx 0.684853$$

$$f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} - 2sin\frac{5\pi}{3}$$

$$= \frac{5\pi}{3} - 2\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{5\pi}{3} + \sqrt{3} \approx 6.968039$$

The values of f(x) at the end points of the intervals are

$f(0) = 0 - 2sin0 = 0$

$f(2\pi) = 2\pi - 2\sin(2\pi) = 2\pi \; = \; 6.28$

Absolute minimum value is $f\left(\frac{\pi}{3}\right) = -0.684$

Absolute maximum value is $f\left(\frac{5\pi}{3}\right) = 6.9680$

**(iii) $f(x) = x - logx$ on $[\frac{1}{2}, 2]$**

**Solution:**

$f(x) = x - logx$ is continuous on $[\frac{1}{2}, 2]$

$$f'(x) = 1 - \frac{1}{x}$$

$$f'(x) = 0 \Rightarrow 1 - \frac{1}{x} = 0$$

$$\Rightarrow \frac{x-1}{x} = 0$$

$\Rightarrow x = 1$ is the critical point.

The value of f(x) at critical point is

$f(1) = 1 - log1 = 1 - 0 \; = 1$

The values of f(x) at the end points of the intervals are

$$f\left(\frac{1}{2}\right) = \frac{1}{2} - log\,\frac{1}{2}$$

$$= \frac{1}{2} - (-0.6931)$$

$$= 1.1931$$

$$f(2) = 2 - \log 2$$

$$= 2 - 0.6931$$

$$= 1.3068$$

Absolute maximum value is f(2) = 1.3068

Absolute minimum value is f(1) = 1

## Rolle's Theorem:

Let f be a function that satisfies the following three conditions:

1) $f$ is continuous on the closed interval $[a, b]$

2) $f$ is differentiable on the open interval $(a, b)$

3) $f(a) = f(b)$

Then there exists a number c in $(a, b)$ such that $f\,'(c) = 0$

**Example:**

**Verify Rolle's theorem for the following functions on the given interval**

**a)** $\boldsymbol{f(x) = x^3 - x^2 + 6x + 2\,,[0, 3]}$

**b)** $\boldsymbol{f(x) = \sqrt{x} - \frac{1}{3}x\,,[0, 9]}$

**Solutions:**

**a)** $\boldsymbol{f(x) = x^3 - x^2 + 6x + 2\,,[0, 3]}$

**Solution:**

$f(x)$ is continuous on $[0, 3]$

$f(x)$ is differentiable on $[0, 3]$

$$f(0) = 2$$

$$f(3) = 27 - 9 + 18 + 2 = 38$$

$$f(0) \neq f(3)$$

Hence the Rolle's theorem is not satisfied.

**b)** $\boldsymbol{f(x) = \sqrt{x} - \frac{1}{3}x\,,[0, 9]}$

**Solution:**

$f(x)$ is continuous on $[0,9]$

$f(x)$ is differentiable on $[0,9]$

$$f(0) = 0$$

$$f(9) = \sqrt{9} - \frac{9}{3} = 3 - 3 = 0$$

$$f(0) = 0 = f(9)$$

$$\Rightarrow f(x) = \sqrt{x} - \frac{x}{3}$$

$$\Rightarrow \ f'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{3}$$

$$\Rightarrow f'(x) = 0 \Rightarrow \frac{1}{2\sqrt{x}} - \frac{1}{3} = 0$$

$$\Rightarrow \frac{1}{2\sqrt{x}} = \frac{1}{3}$$

$$\Rightarrow \sqrt{x} = \frac{3}{2}$$

Squaring, $x = \frac{9}{4} = 2.25 \in (0,9)$

Hence Rolle's theorem is verified.

**Example:**

**Prove that equation $x^3 - 15x + c = 0$ has atmost one real root in the interval [-2 , 2]**

**Solution:**

$$Let \ \ f(x) = x^3 - 15x + c = 0$$

$$f(-2) = -8 + 30 + c = 22 + c$$

$$f(2) = 8 - 30 + c = -22 + c$$

$$f'(x) = 3x^2 - 15$$

Now if there were two points x = a, b such that$f(x) = 0$

∴By Rolle's theorem there exists a point $x = c$ in between them, where $f'(c) = 0$

$$\text{Now } f'(x) = 0 \Rightarrow 3x^2 - 15 = 0$$

$$\Rightarrow x^2 = 5$$

$$\Rightarrow x = \pm\sqrt{5} = \pm 2.236$$

Here both values lies outside [-2, 2]

∴ $f$ has no more than one zero.

⇒ $f(x)$ has exactly one real root.

**Example:**

**Let $f(x) = 1 - x^{2/3}$, Show that $f(-1) = f(1)$but there is no number c in (-1 , 1) such that $f\,'(x) = 0$. Why does this not contradict Rolle's theorem?**

**Solution:**

Given $f(x) = 1 - x^{2/3}$

$$\Rightarrow f(-1) = 1 - (-1)^{\frac{2}{3}} = 0$$

$$\Rightarrow f(1) = 1 - 1^{2/3} = 0$$

$$\therefore f(-1) = f(1)$$

$$\Rightarrow f'(x) = -\frac{2}{3}x^{-1/3}$$

$$\Rightarrow f'(x) = 0 \Rightarrow -\frac{2}{3}x^{-1/3} = 0$$

$$\Rightarrow x^{-1/3} = 0$$

$$\Rightarrow \left(x^{-1/3}\right)^3 = 0^3$$

$$\Rightarrow x^{-1} = 0$$

$$\Rightarrow \frac{1}{x} = 0$$

$$\Rightarrow x = \infty$$

There is no number c in $(-1\,,1)$

f is not differentiable on $(-1\,,1)$

## Increasing/ Decreasing Test

## Definition:

(a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.

(b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

## The first derivative test

## Definition:

**Suppose that c is a critical number of a continuous function $f$.**

(a) If $f$ ‘ changes from positive to negative at c , then $f$ has a local maximum at c.

(b) If $f$ ‘ changes from negative to positive at c , then $f$ has a local minimum at c.

(c) If $f'$ does not change sign at c ( for example if $f'$ is positive on both sides of c or negative on both sides), then $f$ has no local maximum or minimum at c.

### Definition:

If the graph of $f$ lies above all of its tangents on an interval I, then it is called concave upward on I. If the graph of $f$ lies below all of its tangents on an interval I, then it is called concave downward on I.

### Note:

Concave upward $\equiv$ convex downward

Concave downward $\equiv$ convex upward

## Concavity Test

### Definition:

(a) If $f''(x) > 0$ for all x in I, then the graph of $f$ is concave upward on I.

(b) If $f''(x) < 0$ for all x in I, then the graph of $f$ is concave downward on I.

### Definition:

A point P on a curve $y = f(x)$is called an inflection point iff is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P.

## The Second Derivative Test

### Definition:

Suppose $f''$ is continuous near c,

(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

(b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

**Example:**

**Find where the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing and where it is decreasing.**

**Solution:**

$$\text{Given } f(x) = 3x^4 - 4x^3 - 12x^2 + 5$$

$$f'(x) = 12x^3 - 12x^2 - 24x$$

$$= 12x(x^2 - x - 2)$$

$$= 12x(x-2)(x+1)$$

$$f'(x) = 0 \Rightarrow 12x(x-2)(x+1) = 0$$

$$\Rightarrow x(x-2)(x+1) = 0$$

$\Rightarrow x = 0, 2, -1$are the critical values.

We divide the real line into intervals whose end points are the critical points. $x = 0, 2, -1$ and list them in a table

| Interval | $12x$ | $x - 2$ | $x + 1$ | $f'(x)$ | $f(x)$ |
|---|---|---|---|---|---|
| $x < -1$ | - | - | - | - | decreasing |
| $-1 < x < 0$ | - | - | + | + | increasing |
| $0 < x < 2$ | + | - | + | - | decreasing |
| $x > 2$ | + | + | + | + | increasing |

∴The function is increasing in $-1 < x < 0$ and $x > 2$ and it is decreasing in $x < -1$ and $0 < x < 2$

**Example:**

**Find the local maximum and minimum values of $y = x^5 - 5x + 3$ using both the first and second derivative tests.**

**Solution:**

Given $y = f(x) = x^5 - 5x + 3$

$$f'(x) = 5x^4 - 5$$

$$f'(x) = 0 \Rightarrow 5x^4 - 5 = 0$$

$$\Rightarrow x^4 - 1 = 0 \Rightarrow x^4 = 1 \Rightarrow x^2 = \pm 1$$

$\Rightarrow x = 1, -1$are the critical points.

| Interval | Sign of $f'$ | Behaviour of f |
|---|---|---|
| $-\infty < x < 1$ | + | increasing |
| $-1 < x < 1$ | - | decreasing |
| $1 < x < \infty$ | + | increasing |

First derivative test tells us that

(i) Local maximum at $x = -1$

$f(-1) = -1 + 5 + 3 = 7$

Second derivative test tells us that

(ii) Local minimum at $x = 1$

$$f(1) = 1 - 5 + 3 = -1$$

$$f''(x) = 20x^3$$

$f''(x) = 0 \Rightarrow 20x^3 = 0 \Rightarrow x = 0$

| Interval | $f''(x)$ | Behaviour of $f$ |
|---|---|---|
| $(-\infty,0)$ | - | Concave down |
| $(0,\infty)$ | + | Concave up |

$f'(1) = 0$ , $f''(1) = 20$ , $f(1) = -1$ is a local minimum

$f'(-1) = 0$ , $f''(-1) = -20$ , $f(-1) = 7$ is a local maximum

**Example:**

**If $f(x) = 2x^3 + 3x^2 - 36x$ find the intervals on which is increasing or decreasing, the local maximum and local minimum values of f , the intervals of concavity and the inflection points.**

**Solution:**

Given $f(x) = 2x^3 + 3x^2 - 36x$

$$f'(x) = 6x^2 + 6x - 36$$

$$f'(x) = 0 \Rightarrow 6(x^2 + x - 6) = 0$$

$$\Rightarrow 6(x+3)(x-2) = 0$$

$\Rightarrow x = -3$ , 2are the critical points.

$$f''(x) = 12x + 6$$

We divide the real line into intervals whose end points are the critical points $x = 2$ , $-3$ and list them in a table.

| Interval | $6(x+3)$ | $x-2$ | $f'(x)$ | $f(x)$ |
|---|---|---|---|---|
| $x < -3$ | $-$ | $-$ | $+$ | increasing |
| $-3 < x < 2$ | $+$ | $-$ | $-$ | decreasing |
| $x > 2$ | $+$ | $+$ | $+$ | increasing |

Now we apply the first derivative test to find the local extremum values.

$f(x)$changes from increasing to decreasing at $x = -3$ . Thus the function has a local maximum $x = -3$ and local maximum value is $f(-3) = 2(-3)^3 + 3(-3)^2 - 36(-3)$

$$= 2(-27) + 3(9) + 108$$

$$= -54 + 27 + 108 \quad = 81$$

$f(x)$ changes from decreasing to increasing at $x = 2$ . Thus the function has a local minimum $x = 2$ and local minimum value is $f(2) = 2(2)^3 + 3(2)^2 - 36(2)$

$$= 2(8) + 3(4) - 72$$

$$= 16 + 12 - 7 \ = -44$$

For concavity test ,$f''(x) = 0$

$$\Rightarrow 12x + 6 = 0$$

$$\Rightarrow x = -\frac{1}{2}$$

We divide the real line into intervals whose end points are the critical points $x = -\frac{1}{2}$and list them in a table.

| Interval | $f''(x)$ | concavity |
|---|---|---|
| $x < -1/2$ | $-$ | downward |
| $x > -1/2$ | $+$ | upward |

Since the curve changes from concave downward to concave upward at $x = -\frac{1}{2}$

The point of inflection is $[-\frac{1}{2}, f(-\frac{1}{2})]$

$$f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{1}{2}\right)^2 - 36\left(-\frac{1}{2}\right)$$

$$= 2\left(-\frac{1}{8}\right) + 3\left(\frac{1}{4}\right) + 18$$

$$= -\frac{1}{4} + \frac{3}{4} + 18$$

$$= \frac{-1+3+72}{4}$$

$$= \frac{74}{4} \ = \frac{37}{2}$$

Hence the point of inflection are $(-\frac{1}{2}, \frac{37}{2})$

**Example:**

**Find the interval of concavity and the inflection points. Also find the extreme values on what interval is f increasing or decreasing.**

**a)** $(x) = \boldsymbol{sinx + cosx}$ **,** $\boldsymbol{0 \leq x \leq 2\pi}$

**b)** $\boldsymbol{f(x) = e^{2x} + e^{-x}}$

**c)** $\boldsymbol{f(x) = x + 2sinx}$ **,** $\boldsymbol{0 \leq x \leq 2\pi}$

**Solution:**

**a)** $(x) = \boldsymbol{sinx + cosx}$ **,** $\boldsymbol{0 \leq x \leq 2\pi}$

$f'(x) = cosx - sinx$

$f'(x) = 0 \Rightarrow cosx = sinx$

$\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$ are the critical points.

| Interval | Sign of f ' | Behaviour of f |
|---|---|---|
| $0 < x < \frac{\pi}{4}$ | + | increasing |
| $\frac{\pi}{4} < x < \frac{5\pi}{4}$ | + | increasing |
| $\frac{5\pi}{4} < x < 2\pi$ | - | decreasing |

(i) Maximum at $\frac{\pi}{4}$, $f'(\frac{\pi}{4}) = sin\frac{\pi}{4} + cos\frac{\pi}{4}$

$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

(ii) Minimum at $\frac{5\pi}{4}$, $f'(\frac{5\pi}{4}) = sin\frac{5\pi}{4} + cos\frac{5\pi}{4}$

$= -\sqrt{2}$

$f''(x) = -sinx - cosx = -(sinx + cosx)$

$f''(x) = 0 \Rightarrow -(sinx + cosx) = 0$

$\Rightarrow sinx = -cosx$

$\Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$

| Interval | Sign of $f''$ | Behaviour of f |
|---|---|---|

| | | |
|---|---|---|
| $0 < x < \frac{3\pi}{4}$ | $-$ | Concave down |
| $\frac{3\pi}{4} < x < \frac{7\pi}{4}$ | $+$ | Concave up |
| $\frac{3\pi}{4} < x < 2\pi$ | $-$ | Concave down |

Inflection points are $(\frac{3\pi}{4}, 0)$ , $(\frac{7\pi}{4}, 0)$

Since $f\ (\frac{3\pi}{4}) = 0, f\ (\frac{7\pi}{4}) = 0$

b) $f(x) = e^{2x} + e^{-x}$

$$f'(x) = 2e^{2x} - e^{-x}$$

$$f'(x) = 0 \Rightarrow 2e^{2x} - e^{-x} = 0$$

$$\Rightarrow 2e^{2x} = e^{-x}$$

$$\Rightarrow e^{3x} = \frac{1}{2}$$

$$\Rightarrow 3x = \log\left(\frac{1}{2}\right)$$

$$\Rightarrow x = \frac{1}{3}[\log 1 - log2]$$

$$\Rightarrow x = \frac{1}{3}[0 - 0.693]$$

$\Rightarrow -0.23$are the critical points.

| Interval | Sign of $f$ ‘ | Behaviour of f |
|---|---|---|
| $-\infty < x < -0.23$ | $-$ | decreasing |
| $-0.23 < x < \infty$ | $+$ | increasing |

The first derivative test tells us that there is a local minimum at $x = -0.23$

$$f(-0.23) = f\left(-\frac{1}{3}log2\right) = f\left(log2^{-\frac{1}{3}}\right)$$

$$= e^{2log2^{-1/3}} + e^{-log2^{-1/3}}$$

$$= e^{log(2^{-1/3})^2} + e^{log(2^{-1/3})^{-1}}$$

$$= (2^{-1/3})^2 + (2^{-1/3})^{-1}$$

$$= (2)^{-2/3} + (2)^{1/3}$$

$$f''(x) = 4e^{2x} + e^{-x}$$

$$f''(x) = 0 \Rightarrow 4e^{2x} + e^{-x} = 0$$

$$\Rightarrow 4e^{2x} = -e^{-x}$$

$$\Rightarrow e^{3x} = -\frac{1}{4}$$

$$\Rightarrow 3x = log\,(-\frac{1}{4})$$

$$\Rightarrow x = \frac{1}{3}\,log\,(-\frac{1}{4})$$

$$\Rightarrow x = \frac{1}{3}(-log\,4\,)$$

$$= -\frac{1}{3}(log\,4\,) \;= -0.46$$

| Interval | Sign of $f''$ | Behaviour of f |
|---|---|---|
| $-\infty < x < -0.46$ | $+$ | Concave up |
| $-0.46 < x < \infty$ | $+$ | Concave up |

No inflection points.

c) $f(x) = x + 2sinx\,,\;\; 0 \le x \le 2\pi$

$$f'(x) = 1 + 2cosx$$

$$f'(x) = 0 \Rightarrow 2cosx = -1$$

$$\Rightarrow cosx = -\frac{1}{2}$$

$\Rightarrow x = \frac{2\pi}{3}, \frac{4\pi}{3}$ are the critical points.

| Interval | Sign of $f'$ | Behaviour of f |
|---|---|---|
| $0 < x < \frac{2\pi}{3}$ | $+$ | increasing |
| $\frac{2\pi}{3} < x < \frac{4\pi}{3}$ | $-$ | decreasing |
| $\frac{4\pi}{3} < x < 2\pi$ | $+$ | increasing |

The first derivatives test tells us that there is a

(i) Local maximum at $\frac{2\pi}{3}$

$$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\sin\left(\frac{2\pi}{3}\right) = 3.83$$

(ii) Local minimum at $\frac{4\pi}{3}$

$$f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + 2\sin\left(\frac{4\pi}{3}\right) = 2.46$$

$$f''(x) = -2sinx$$

$$f''(x) = 0 \Rightarrow -2sinx = 0$$

$$\Rightarrow sinx = 0 \Rightarrow x = 0, \pi, 2\pi$$

| Interval | Sign of $f''$ | Behaviour of f |
|---|---|---|
| $0 < x < \pi$ | $+$ | Concave up |
| $\pi < x < 2\pi$ | $-$ | Concave down |

Inflection points are $(\pi, \pi)$